25901239 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Value Addition Course (VAC) - MG4VACECT200, MG4VACECC200

- ENVIRONMENTAL MONITORING USING SENSORS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Floods and droughts are linked to [U] / [CO1]

   a). Stable climate b). Magnetic fields

   c). Reduced evaporation d). Climate variability

2. Primary pollutants are [U] / [CO1]

   a). Formed by reaction b). Found in soil

   c). Produced in water d). None the of above

3. AQI is usually reported using [K] / [CO1]

   a). Alphabetical order b). Map symbols only

   c). A numerical scale d). Sound signals

4. AQI-based planning includes [U] / [CO1]

   a). Increasing smoke zones b). Creating green spaces

c). Removing trees d). Reducing monitoring

5. Chemical parameter monitoring ensures: [U] / [CO1]

a). Safe air and water quality b). Faster plant growth

c). More rainfall d). Less sunlight

6. Using public transport helps in [U] / [CO1]

a). Reducing air pollution b). Increasing noise

c). Producing waste d). Soil erosion

7. Global warming is caused by [K] / [CO1]

a). Increase in greenhouse gases b). Decrease in sunlight

c). Ocean currents d). Soil erosion

8. Pollution control measures are based on [U] / [CO1]

a). Financial b). Monitoring

c). Historical d). Political

9. Environmental monitoring means [K] / [CO1]

a). Systematic observation of environment b). Industrial production

c). Medical treatment d). Space research

10. Without the greenhouse effect [K] / [CO1]

a). Earth would be hotter b). Earth would become very cold

c). No sunlight would reach d). Seasons would stop

11. The natural greenhouse effect is [U] / [CO1]

a). Harmful to all organisms b). Essential for life on Earth

c). Useless process d). Only industrial

12. One limitation of MQ-135 is [K] / [CO2]

a). Low availability b). Cross-sensitivity to gases

c). Difficult to use d). Large size

13. Which of the following is an example of a gas sensor? [U] / [CO2]

a). Thermistor b). MQ-135

c). LDR d). Ultrasonic sensor

14. MQ135 sensor works based on change in [K] / [CO2]

a). Color b). Sound

c). Electrical resistance d). Light intensity

15. Environmental sensors help to detect [U] / [CO2]

a). Pollution b). Internet speed

c). Screen brightness d). Battery level

16. Methane gas is dangerous mainly due to [U] / [CO2]

a). Its color b). Its smell

c). Its flammable nature d). Its weight

17. Early smoke detection helps in [K] / [CO2]

a). Delaying response b). Saving lives and property

c). Increasing damage d). Ignoring hazards

18. Humidity sensors are used to measure [K] / [CO2]

a). Temperature b). Pressure

c). Moisture in air d). Light

19. Chemical sensors are used to detect [U] / [CO2]

a). Light b). Sound

c). Chemical substances d). Motion

20. The digital output (DO) pin of MQ135 provide [K] / [CO2]

a). Continuous values b). ON/OFF signal

c). Temperature value d). Humidity value

21. Thermocouple works on the principle of [U] / [CO2]

a). Photoelectric effect b). Seebeck effect

c). Hall effect d). Piezoelectric effect

22. Carbon monoxide (CO) is a [U] / [CO2]

a). Colorless and odorless gas b). Green gas

c). Liquid fuel d). Solid material

23. MQ135–Arduino system is commonly used in [K] / [CO2]

a). Air quality monitoring systems b). Video games

c). Printers d). Washing machines

24. A city wants to monitor rainfall and temperature automatically. Which system should be implemented? [A] / [CO3, CO4]

a). IoT-based weather monitoring system b). Smart TV system

c). Online banking system d). Library management system

25. How do industries contribute to air pollution? [U] / [CO4]

a). By cleaning the environment b). By planting trees

c). By releasing harmful gases and smoke d). By producing oxygen

26. Which type of weather station is most suitable for temporary weather monitoring during outdoor sports events? [A] / [CO3, CO4]

a). Manual b). Remote

c). Portable d). Automatic

27. Which combination of parameters would best help design a flood warning system? [C] / [CO3, CO4]

a). Temperature and wind speed b). Humidity and pressure

c). Temperature and humidity d). Rainfall, wind speed, and atmospheric pressure

28. If a vehicle shows high Nitrous oxide emissions, which system is likely malfunctioning? [An] / [CO4]

a). Headlight system
b). Steering system
c). Air conditioning system
d). Exhaust Gas Recirculation (EGR) system

29. Environmental education is usually provided in: [An] / [CO4]

a). Shopping malls
b). Factories only
c). Schools and colleges
d). Railway stations

30. Why are electrostatic precipitators suitable for industries? [U] / [CO4]

a). They increase smoke production
b). They reduce oxygen levels
c). They efficiently remove fine particles from exhaust gases
d). They consume no electricity

31. Which measure can a government apply to reduce vehicle pollution in cities? [A] / [CO4]

a). Remove emission testing
b). Enforce strict vehicle emission standards
c). Increase fuel subsidies
d). Reduce public transport services

32. Which of the following is a main component of a weather station? [K] / [CO3, CO4]

a). Speaker
b). Printer
c). Sensors
d). Keyboard

33. Which of the following is commonly monitored using IoT in the environment? [K] / [CO3, CO4]

a). Air quality
b). Television channels
c). Movie ratings
d). Social media posts

34. Automatic weather stations are preferred because they: [A] / [CO3, CO4]

a). Work only in daylight
b). Require constant human observation
c). Provide continuous real-time data
d). Are used only in laboratories

35. Solar power is commonly used in weather stations because it: [U] / [CO3, CO4]

a). Improves sensor sensitivity

b). Reduces the number of sensors

c). Provides reliable power in remote areas

d). Replaces the data logger

[1x35 = 35]

Part B

Short Essay Questions

Answer any **3** questions

Each question carries **5** marks

36. How do greenhouse gases disturb natural ecological balance [U] / [CO1]

37. Explain the need for environmental monitoring. [U] / [CO1]

38. What is MQ gas sensor series? [K] / [CO2]

39. How does ozone affect the environment? [K] / [CO2]

40. State one importance of IoT in daily life. [C] / [CO3, CO4]

[5x3 = 15]